

Positioning Error

The formal, predicted and measured accuracy parameters are discussed in this article. The first one is a measure of the uncertainty of the estimates, according to the statistical characterisation of the errors and the linear model used for the position estimate.

The predicted accuracy, provides the expected position accuracy based on a simple statistical description of the measurement errors. Its computation does not require the measurements, but only its standard deviation σ , and the approximated satellites and user location coordinates. Thence, it can be computed at any point and at any time with the almanac, without needing the measurements.

The measured accuracy is the measure of the true error and must be assessed with the actual measurements.

Fundamentals

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Author(s)	J. Sanz Subirana, J.M. Juan Zornoza and M. Hernández-Pajares, Technical University of Catalonia, Spain.
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Formal Accuracy

Denoting as P_{xx} , P_{yy} , P_{zz} , P_{tt} the diagonal elements of matrix $\mathbf{P}_{\Delta x}$ of (see [Best Linear Unbiased Minimum-Variance Estimator \(BLUE\)](#))

$$\mathbf{P} = (\mathbf{G}^T \mathbf{R}^{-1} \mathbf{G})^{-1} \quad (1)$$

we call formal errors of the estimated components of vector $\Delta X = (dx, dy, dz, dt)$ to the standard deviations:

$$\sigma_X = \sqrt{P_{xx}} \quad \sigma_Y = \sqrt{P_{yy}} \quad \sigma_Z = \sqrt{P_{zz}} \quad \sigma_T = \sqrt{P_{tt}} \quad (2)$$

The previous expressions provides a characterisation of the quality of the coordinates and clock estimates (linked to the measurement error model assumed). Thence, they are not the actual error, but only a measure of the uncertainty of the error estimates.

Previous expressions give the errors in the ECEF XYZ coordinates. Nevertheless, it is usually more meaningful to a user to think in terms of horizontal and vertical position error, or East, North and Up (ENU) coordinates.

Let $\mathbf{R}$ the transformation matrix (3) of ENU coordinates to XYZ, (i.e., which columns are the unit orthogonal vectors $\{\hat{\mathbf{e}}, \hat{\mathbf{n}}, \hat{\mathbf{u}}\}$ as expressed in the XYZ coordinates system [\[footnotes 1\]](#) at a point of longitude λ and latitude φ) (see [Transformations between ECEF and ENU coordinates](#)):

$$\mathbf{R} = \begin{bmatrix} -\sin \lambda & -\sin \varphi \cos \lambda & \cos \varphi \cos \lambda \\ \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \sin \lambda \\ 0 & \cos \varphi & \sin \varphi \end{bmatrix} \quad (3)$$

thence, $\mathbf{P}_{enu} = \mathbf{R}^T \mathbf{P}_{xyz} \mathbf{R}$ is given, where $\mathbf{P}_{xyz}$ is the sub-matrix of $\mathbf{P}_{\Delta X}$ containing solely the geometric components [\[footnotes 2\]](#). From this covariance matrix, they can be defined:

$$\sigma_{East} = \sqrt{P_{ee}} \quad \sigma_{North} = \sqrt{P_{nn}} \quad \sigma_{Vertical} = \sqrt{P_{uu}} \quad (4)$$

For the horizontal error, there are usually also defined:

$$\sigma_{Horizontal} = \sqrt{P_{ee} + P_{nn}} \quad (5)$$

It is usual to express the Horizontal error in two main directions where the covariance matrix $\mathbf{P}_{enu}$ is diagonal. They define two orthogonal axis, associated to the minor and major axis of the error ellipse. The major axis is given by:

$$\sigma_{H_{major}} = \sqrt{\frac{P_{ee} + P_{nn}}{2} + \sqrt{\left(\frac{P_{ee} - P_{nn}}{2}\right)^2 + P_{en}^2}} \quad (6)$$

Predicted Accuracy: Dilution of Precision

Assuming the simple statistical model:

$$\mathbf{m} = E[\boldsymbol{\varepsilon}] = 0 \quad ; \quad \mathbf{R} = E[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T] = \sigma^2 \mathbf{I} \quad (7)$$

the covariance matrix of $\Delta \mathbf{X}$ estimate is given by (see [Parameters adjustment](#))

$$\mathbf{P}_{\Delta X} = \sigma^2 (\mathbf{G}^T \mathbf{G})^{-1} \quad (8)$$

This expression depends upon two factors: 1) the measurements (prefit-residuals) variance σ^2 , and 2) the geometry matrix ($\mathbf{G}$), which is linked only to the receiver-satellite rays geometry, (see [Code Based Positioning \(SPS\)](#)).

The Root Mean Square Error (RMSE) is given by:

$$\text{RMSE}_{\Delta X} = \sqrt{\text{trace}(\mathbf{P}_{\Delta X})} = \sigma \sqrt{\text{trace}[(\mathbf{G}^T \mathbf{G})^{-1}]} \quad (9)$$

what means that, the trace of matrix $(\mathbf{G}^T \mathbf{G})^{-1}$ is a scale factor on σ for the $\text{RMSE}_{\Delta X}$.

Notice that, as the matrix $\mathbf{G}$ does not depend on the measurements, but only on the geometry, it can be computed from the almanac (because not accurate satellite positions are needed), i.e., does not require receiver measurements.

On the basis of this simple approach, the following Dilution of Precision (DOP) parameters are defined:

$$\mathbf{Q}_{\Delta X} \equiv (\mathbf{G}^T \mathbf{G})^{-1} = \begin{bmatrix} q_{xx} & q_{xy} & q_{xz} & q_{xt} \\ q_{xy} & q_{yy} & q_{yz} & q_{yt} \\ q_{xz} & q_{yz} & q_{zz} & q_{zt} \\ q_{xt} & q_{yt} & q_{zt} & q_{tt} \end{bmatrix} \quad (10)$$

■ **Geometric Dilution of Precision:**

$$GDOP = \sqrt{q_{xx} + q_{yy} + q_{zz} + q_{tt}}$$

■ **Position Dilution of Precision:**

$$PDOP = \sqrt{q_{xx} + q_{yy} + q_{zz}}$$

■ **Time Dilution of Precision:**

$$TDOP = \sqrt{q_{tt}}$$

Using (3) the sub-matrix matrix $\mathbf{Q}_{xyz}$ of $\mathbf{Q}_{\Delta X}$ can be transformed to ENU coordinates as $\mathbf{Q}_{enu} = \mathbf{R}^T \mathbf{Q}_{xyz} \mathbf{R}$, in order to define:

■ **Horizontal Dilution of Precision:**

$$HDOP = \sqrt{q_{ee} + q_{nn}}$$

■ **Vertical Dilution of Precision:**

$$VDOP = \sqrt{q_{uu}}$$

Thence, estimations of the expected accuracy are given by:

- $GDOP, \sigma_0$... geometric precision in position and time
- $PDOP, \sigma_0$... precision in position
- $TDOP, \sigma_0$... precision in time
- $HDOP, \sigma_0$... precision in horizontal positioning
- $VDOP, \sigma_0$... precision in vertical positioning

where, basically, DOP represents an approximate ratio factor between the precision in the measurements (σ_0) and in positioning. This ratio is computed only from the satellites-receiver geometry.

Measured Accuracy

Let ΔE_i , ΔN_i and ΔU_i the errors in the East, North and Vertical components of the i -th position estimate sample. The RMS vertical, horizontal (2-D) and 3-D errors are defined as ^[footnotes 3]:

$$\text{rms vertical error} = \sqrt{\frac{1}{n} \sum_{i=1}^n \Delta U_i^2} \quad (11)$$

$$\text{2-D rms horizontal error} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\Delta E_i^2 + \Delta N_i^2)} \quad (12)$$

$$\text{3-D rms error} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\Delta E_i^2 + \Delta N_i^2 + \Delta U_i^2)} \quad (13)$$

Other measures of the quality of the position estimates are:

- 50th or 95th percentiles of horizontal, vertical and 3-D errors.
- CEP: Circular Error Probable, as the 50th percentile of horizontal error.
- SEP: Spherical Error Probable, as the 50th percentile of 3-D error.
- 2drms: calculated as twice the 2-D rms horizontal error given by (12).

Assuming that the position estimates follows a multivariate normal distribution centred at the true position and the errors in north, east and up are uncorrelated, with $\sigma_E = \sigma_N$ and $\sigma_U = 2(\sigma_e^2 + \sigma_N^2)^{1/2}$, then, the following relations can be considered ([Misra and Enge, 2001] ^[1]):

$$\begin{aligned} \text{rms vertical error} &\simeq 0.5 \times \text{vertical error (95\%)} \\ &\simeq 2\text{drms} \\ &0.9 \times \text{3-D rms error} \\ \text{2-D rms error} &= 0.5 \times 2\text{drms} \\ &\simeq 0.6 \times \text{horizontal error (95\%)} \\ &1.2 \times \text{CEP} \\ \text{3-D rms error} &= 2.2 \times \text{2-D rms error} \\ &1.2 \times \text{horizontal error (95\%)} \\ &1.3 \times \text{SEP} \end{aligned}$$

A wide discussion on such accuracy equivalences can be found in [Diggelen, 2007] [2].

Notes

1. ^ Note (λ, φ) are the ellipsoidal coordinates and, thence, the vector $\mathbf{u}$ is orthogonal to the tangent plane to the ellipsoid, which is defined by $(\hat{\mathbf{e}}, \hat{\mathbf{n}})$. If (λ, φ) are taken as the spherical latitude and longitude, thence, the vector $\mathbf{u}$ is in the radial direction and $(\hat{\mathbf{e}}, \hat{\mathbf{n}})$ defines the tangent plane to the sphere.
2. ^ The matrix $\mathbf{P}_{enu}$ can be obtained directly in the ENU coordinates, using the geometry matrix $\mathbf{G}$ computed in the ENU coordinates. In this local system the rows of matrix $\mathbf{G}$ (see Code Based Positioning (SPS)) are $[\cos el^i \sin az^i, \cos el^i \cos az^i, \sin el^i, 1]$, where el^i and az^i are the elevation and azimuth angle of the satellite i observed from the receiver position, (see Transformations between ECEF and ENU coordinates).
3. ^ Notice that the rms equals to the standard derivation s only when the mean position error is zero. Indeed, let

$$m = \frac{1}{n} \sum_{i=1}^n x_i ; rms = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} ; s = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - m)^2} \quad \text{thence} \quad rms^2 = s^2 + m^2$$

References

1. ^ [Misra and Enge, 2001] Misra, P. and Enge, P., 2001. Global Positioning System. Signals, Measurements and Performance.. Ganga-Jamuna Press, Massachusetts, U.S.
 2. ^ [Diggelen, 2007] Diggelen, F. v., 2007. GNSS Accuracy: Lies, Damn Lies, and Statistics. GPS World. 18(1), pp. 26-32.
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